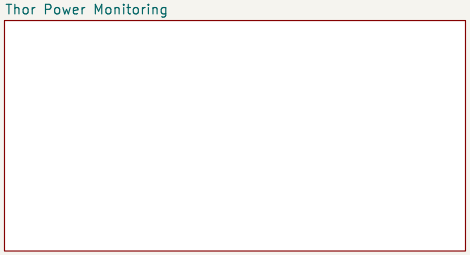
Thor Power Board — Monitoring, Protection and MCU

Sheets connect through global labels. Nets owned by other blocks:

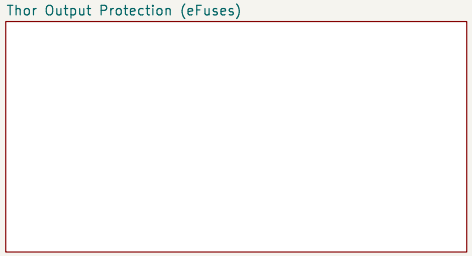
5V_REG, 12V_REG = regulator outputs (into the shunts) | 5V_USB_OUT, 5V_6A_OUT, 12V_1A_OUT = board outputs
5V_REG_EN, 5V_REG_PG = 5V regulator | CHARGER_SELECT/QON/INT/ENABLE = charger (charger I2C shares I2C_SDA/SCL)

Thor Power Monitoring



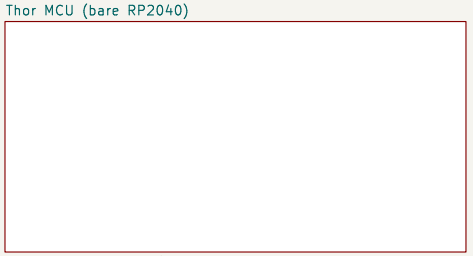
File: Thor_Monitor.kicad_sch

Thor Output Protection (eFuses)



File: Thor_Protection.kicad_sch

Thor MCU (bare RP2040)



File: Thor_MCU_RP2040.kicad_sch

Root sheet
HABET — Iowa State

Sheet: /
File: Thor_Power_System.kicad_sch

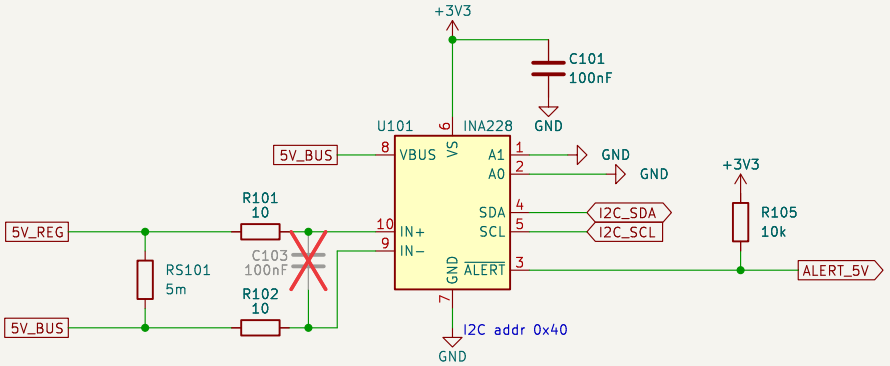
Title: Thor Power Board — Monitoring, Protection, MCU

Size: A4	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

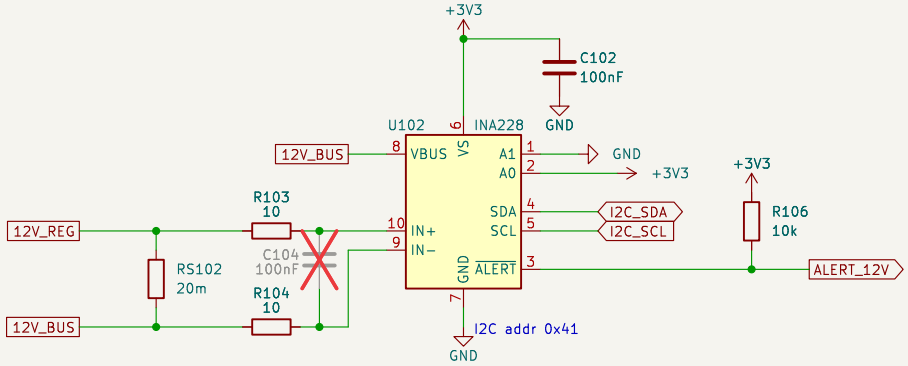
Power Monitoring — 2x INA228

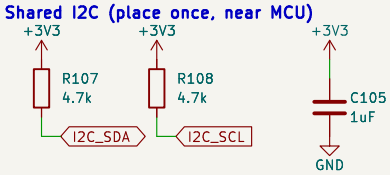
High-side sensing after each regulator, before the eFuses. VS = +3V3 from the Pico (logic supply, not the measured rail).

CH1 — 5V bus (feeds 5V 3A USB + 5V 6A), up to 9A



CH2 — 12V 1A rail





INA228 x2: CH1 5V bus (0x40), CH2 12V rail (0x41)

HABET — Iowa State

Sheet: /Thor Power Monitoring/
File: Thor_Monitor.kicad_sch

Title: Thor Power Monitoring

Size: A4	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

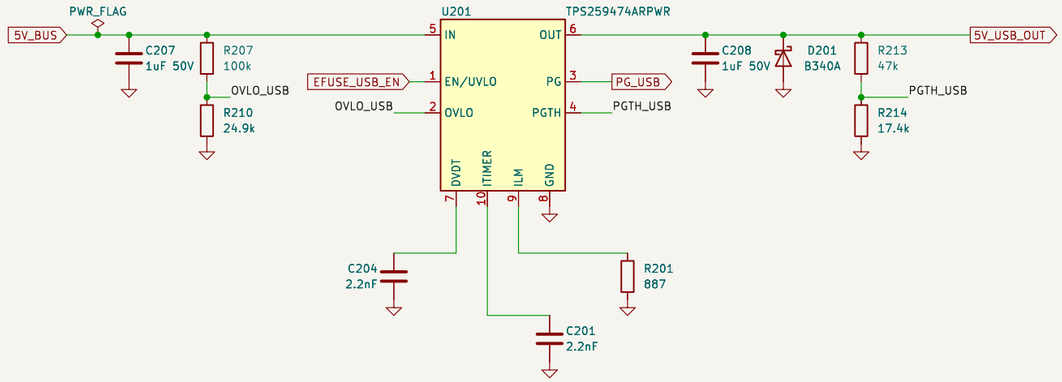
Output Protection — 3x TI eFuse (resettable)

EN pins default ON via 10k pull-ups on the MCU sheet (10k beats the RP2040 50–80k boot pull-down); MCU pulls EN low to turn a rail off or reset a fault.

OVLO dividers cut the output if the input is too high (5V rails: 6.0V, 12V rail: 14.1V). Unlabelled triangles = GND.

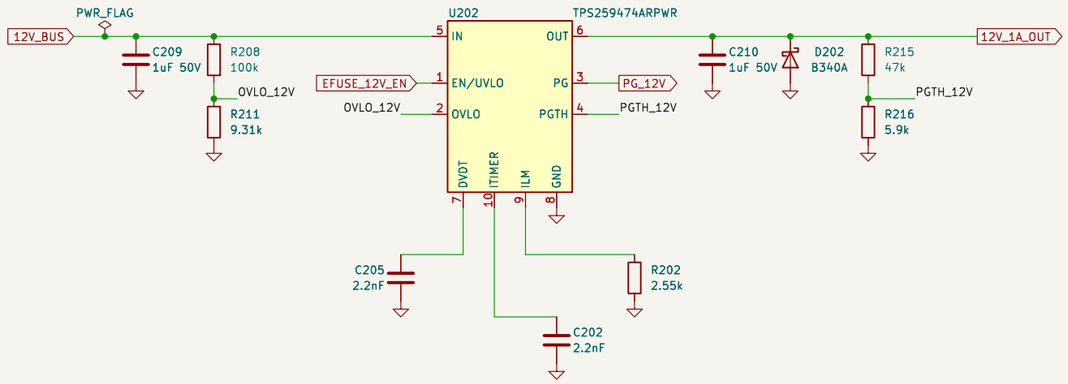
U201 — 5V 3A USB output (also replaces the 'Switch' block)

TPS259474A: circuit breaker, auto-retry 110ms | ILIM = 3334/887 = 3.76A | OVLO 6.02V | PG asserts at 4.44V



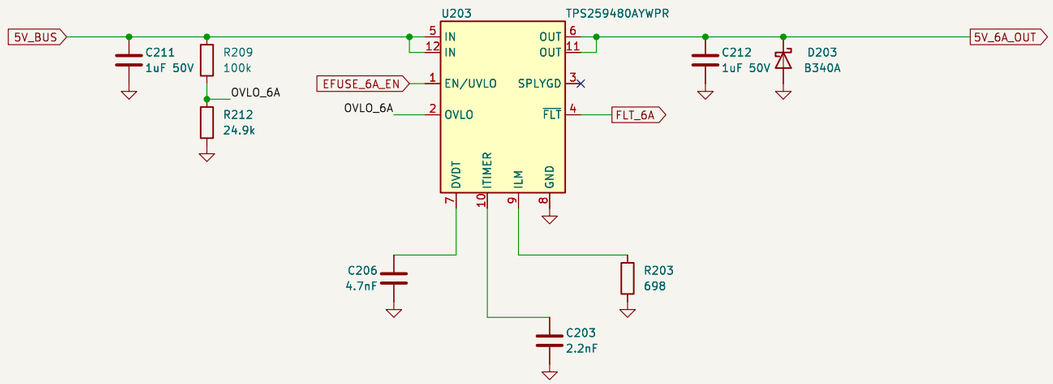
U202 — 12V 1A output

TPS259474A: circuit breaker, auto-retry 110ms | ILIM = 3334/2550 = 1.31A | OVLO 14.1V | PG asserts at 10.76V



U203 — 5V 6A output

TPS259480A: active current limit, auto-retry 103ms | ILIM = 4834/698 = 6.93A | OVLO 6.0V | FLT is active-low



U201 5V USB 3.76A | U202 12V 1.31A | U203 5V 6.93A

HABET — Iowa State

Sheet: /Thor Output Protection (eFuses)/

File: Thor_Protection.kicad_sch

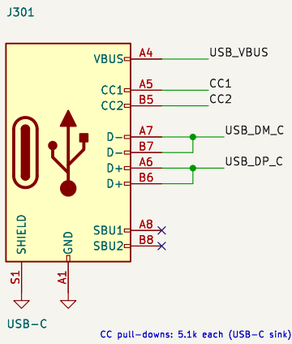
Title: Thor Output Protection (eFuses)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

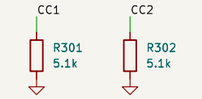
MCU — bare RP2040 (replaces the Pico module)

Based on Raspberry Pi 'Hardware design with RP2040' minimal design. Parts chosen for −40C.
Net names + GPIO numbers match the merged Pico sheet (I2C on the valid I2C0 pair GP4/GP5).

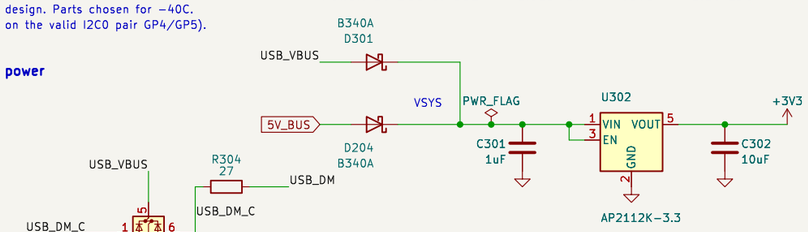
USB-C: programming (drag-and-drop UF2) + bench power

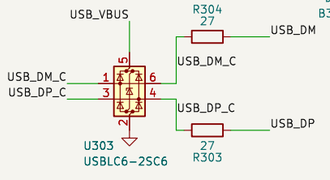


CC pull-downs: 5.1k each (USB-C sink)



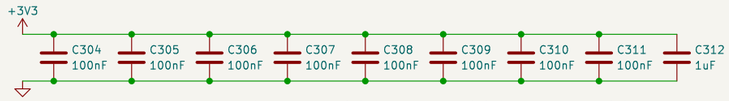
Power: 5V_BUS or USB -> diode-OR -> VSYS -> 3.3V LDO



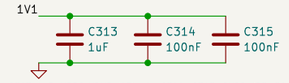


27R series resistors: place close to the RP2040

Decoupling — place each cap right at its pin

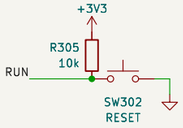


C304–C309: IOVDD pins 1, 10, 22, 33, 42, 49 | C310: USB_VDD (48) | C311: ADC_AVDD (43) | C312 1uF: VREG_IN (44)



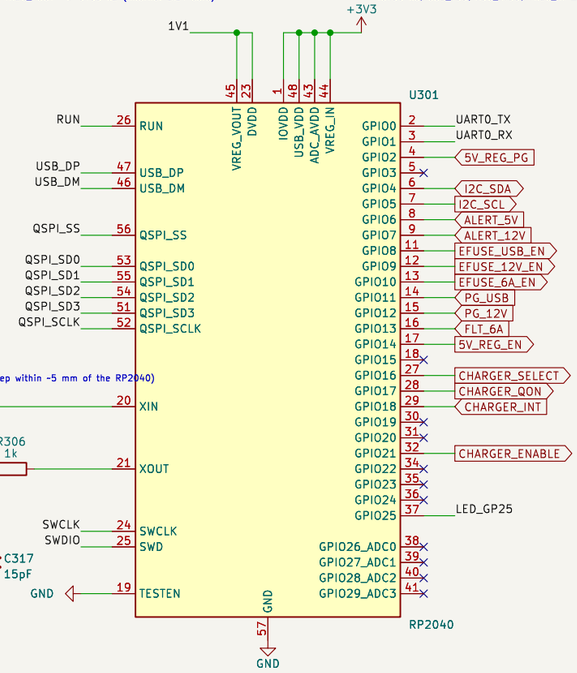
C313 1uF: VREG_VOUT (45) | C314, C315: DVDD (23, 50)



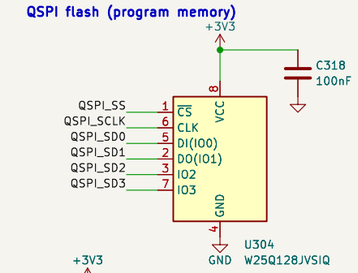


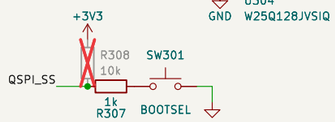
VREG_VOUT -> DVDD x2 (internal 1.1V core)

IOVDD x6, USB_VDD, ADC_AVDD, VREG_IN = +3V3

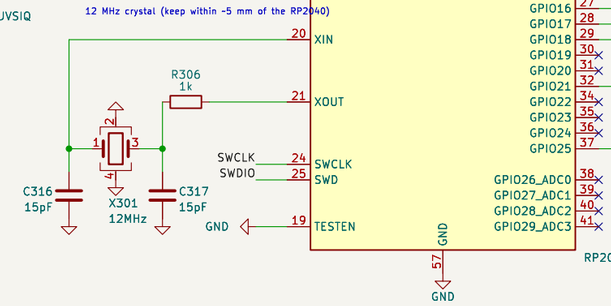


QSPI flash (program memory)

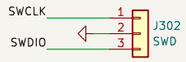




12 MHz crystal (keep within ~5 mm of the RP2040)



Debug (SWD) — RPi Debug Probe order: SWCLK, GND, SWDIO



UART0 debug console (3.3V logic)



GPIO map (same GPIO numbers as the Pico sheet)

GP0/GP1 UART0 TX/RX -> J303 debug header
GP2 5V_REG_PG (pull-up to 3.3V on regulator sheet)
GP4/GP5 I2C0 SDA/SCL: INA228 0x40/0x41 + charger
GP6/GP7 ALERT_5V / ALERT_12V (active low)
GP8–GP10 EFUSE_USB_EN / EFUSE_12V_EN / EFUSE_6A_EN
GP11/GP12 PG_USB / PG_12V (high = good)
GP13 FLT_6A (low = fault)
GP14 5V_REG_EN (must default ON on reg sheet)
GP16 CHARGER_SELECT GP17 CHARGER_QON
GP18 CHARGER_INT GP21 CHARGER_ENABLE
GP25 status LED
Free: GP3, GP15, GP19, GP20, GP22–GP24, GP26–GP29
(GP26–GP29 are ADC inputs)

Firmware: disable RP2040 internal pull-downs on GP6–GP13 at boot.
Caution: driving 5V_REG_EN low turns off the 5V buck, which also powers this MCU.

Status LED on GP25 (same as the Pico's)



Pull-ups (EN default ON = rails power up even if the MCU is not running)



Bare RP2040 — rev B MCU sheet (replaces Pico module)
HABET — Iowa State
Sheet: /Thor MCU (bare RP2040)/
File: Thor_MCU_RP2040.kicad_sch
Title: Thor MCU (bare RP2040)

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/4